Project Report on
CUSTOMER BEHAVIOR ANALYSIS

PROJECT REPORT

Submitted as part of the Data Analytics Internship Program at Alfido Tech

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ABSTRACT

In the rapidly growing e-commerce industry, understanding customer behavior is essential for improving business performance, increasing customer satisfaction, and developing effective customer retention strategies. The Customer Behavior Analysis project focuses on analyzing customer purchasing patterns, transaction behavior, revenue generation, and customer churn using data analytics techniques.

The project follows a structured data analytics workflow that includes Data collection, Data cleaning, Exploratory Data Analysis (EDA), Feature Engineering, SQL-based analysis, statistical analysis, and interactive dashboard development. The customer transaction dataset is processed using Python libraries such as Pandas and NumPy, while SQL is used to perform structured queries and extract meaningful information from the data. Power BI is utilized to transform the analyzed data into an interactive dashboard containing key performance indicators (KPIs), charts, and visualizations.

The analysis focuses on important business metrics such as Total Customers, Total Orders, Total Revenue, and Churn Rate, along with customer purchasing patterns and relationships between different variables. Various visualization techniques, including Bar charts, Area charts, Scatter plots, and KPI cards, are used to present the results in an understandable and interactive manner.

The project aims to convert raw customer data into meaningful business insights that can help identify valuable customer segments, understand purchasing behavior, recognize potential churn patterns, and support data-driven decision-making. The final outcome is an integrated analytical solution that demonstrates how data analytics, SQL, Python, and business intelligence tools can be combined to analyze customer behavior and support effective business strategies.

Keywords: Customer Behavior Analysis, Data Analytics, E-commerce, Customer Segmentation, Customer Churn, Exploratory Data Analysis, SQL, Python, Power BI, Business Intelligence.

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CHAPTER 1

INTRODUCTION

1.1 Background

In the modern e-commerce environment, businesses generate a large amount of customer and transaction data through online purchases, payment activities, product interactions, and customer engagements. This data contains valuable information about how customers behave, what products they purchase, how frequently they make purchases, and which factors may influence their relationship with a business.

Customer Behavior Analysis is the process of examining customer-related data to identify meaningful patterns, trends, and relationships. Analyzing customer behavior enables organizations to understand their customers more effectively and make data-driven business decisions. It can help businesses identify valuable customers, understand purchasing patterns, measure revenue contribution, and identify customers who may be at risk of leaving.

The project **"Customer Behavior Analysis"** focuses on applying data analytics techniques to customer and transaction data to extract meaningful business information. The project follows a structured analytical workflow involving data preprocessing, exploratory data analysis, SQL-based analysis, and business intelligence visualization.

Python is used extensively for data handling, preprocessing, exploratory analysis, and analytical operations using the required Python libraries. After the data is prepared, SQL is used for structured analysis, particularly for **customer churn analysis and deriving business insights**. The analyzed information is then presented through an interactive **Power BI dashboard**, allowing important metrics and patterns to be understood more easily.

The developed dashboard provides a consolidated view of important customer and business metrics such as **Total Customers, Total Revenue, Total Orders, and Churn Rate**, along with different visualizations for understanding customer demographics, purchasing behavior, revenue patterns, and churn-related information.

1.2 Project Overview

The Customer Behavior Analysis project is designed as an end-to-end data analytics solution for analyzing customer and transaction-related information. The project integrates multiple technologies to transform raw data into meaningful and actionable information.

The project begins with the examination and preparation of the dataset. Data quality is checked by identifying missing values, duplicate records, incorrect data types.

1.3 Need for Customer Behavior Analysis

Understanding customer behavior is important for businesses because customer preferences and purchasing patterns directly influence business performance. Simply collecting customer data is not sufficient; organizations need to analyze that data to discover useful patterns.

Customer Behavior Analysis is required for the following reasons:

- **Understanding purchasing patterns:** It helps identify how frequently customers purchase products and how their purchasing behavior varies.
- **Identifying valuable customers:** Analysis can help recognize customers who contribute significantly to overall revenue.
- **Understanding customer segments:** Customers can be studied according to characteristics such as age group, gender, purchasing behavior, and other relevant attributes.
- **Analyzing customer churn:** Identifying churn patterns helps businesses understand which customers may discontinue their relationship with the business.
- **Improving business decisions:** Data-driven insights provide a stronger basis for planning marketing, sales, and customer-retention strategies.
- **Monitoring business performance:** KPIs such as total customers, revenue, orders, and churn rate provide a high-level view of business performance.
- **Improving customer retention:** Understanding churn-related behavior can help businesses develop strategies to retain existing customers.

1.4 Problem Statement

E-commerce businesses generate substantial amounts of customer and transaction data, but raw data alone does not provide clear information about customer behavior or business performance. Without systematic analysis, it can be difficult to identify important purchasing patterns, understand customer segments, determine churn behavior, and extract meaningful business insights.

Therefore, there is a need for an integrated data analytics solution that can process customer data, perform exploratory analysis, use SQL to analyze customer churn and business-related patterns, and present the resulting information through an interactive visualization platform.

The problem addressed by this project is:

To develop a data analytics solution that analyzes customer and transaction data to understand customer behavior, identify churn patterns, derive meaningful business insights using SQL, and present the results through an interactive Power BI dashboard.

1.5 Objectives of the Project

The major objectives of the Customer Behavior Analysis project are:

1. To collect and examine customer and transaction-related data.
2. To clean and preprocess the dataset using Python and the required Python libraries.
3. To perform Exploratory Data Analysis (EDA) to identify patterns, trends, and relationships within the data.
4. To analyze customer purchasing behavior and relevant customer characteristics.
5. To use SQL for structured customer and transaction analysis.
6. To perform **customer churn analysis using SQL**.
7. To derive meaningful **business insights using SQL-based analysis**.
8. To calculate and analyze important business KPIs such as total customers, total revenue, total orders, and churn rate.
9. To develop an interactive Power BI dashboard for visualizing the analytical results.
10. To present customer behavior and business performance in a clear and understandable manner.
11. To support data-driven decision-making through meaningful analytical insights.

1.6 Scope of the Project

The scope of this project covers the complete process of analyzing customer behavior from raw data to business intelligence visualization. The project includes data preprocessing, exploratory analysis, SQL-based analysis, churn analysis, business insight generation, and dashboard development.

The major areas covered by the project include:

- Customer and transaction data analysis
- Data cleaning and preprocessing
- Exploratory Data Analysis
- Customer purchasing behavior
- Customer segmentation and demographic analysis
- Revenue and order analysis
- Customer churn analysis using SQL
- Business insight generation using SQL
- KPI analysis
- Interactive Power BI visualization
- Data-driven business recommendations

The project demonstrates how multiple data analytics technologies can work together to convert raw customer data into meaningful information. The resulting dashboard provides an interactive way to explore important customer and business metrics and supports the interpretation of analytical results.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Customer behavior analysis has become an important application of data analytics in the e-commerce industry. Researchers and businesses use customer data to understand purchasing patterns, customer segments, revenue contribution, and customer retention.

2.2 Customer Behavior Analysis

Customer behavior analysis involves studying customer interactions and purchasing activities to identify patterns and trends. It helps businesses understand customer preferences and make better data-driven decisions.

2.3 Customer Segmentation

Customer segmentation divides customers into groups based on common characteristics such as demographics, purchasing behavior, and transaction patterns. It helps businesses develop targeted strategies for different customer groups.

2.4 Customer Churn Analysis

Customer churn refers to customers who stop purchasing from or interacting with a business. Churn analysis helps identify patterns associated with customer loss and supports customer-retention strategies. In this project, churn analysis is performed using **SQL**.

2.5 Role of Data Analytics in E-Commerce

Data analytics helps e-commerce businesses analyze large volumes of customer and transaction data. It can provide insights into sales performance, customer behavior, purchasing trends, and churn.

2.6 Business Intelligence and Data Visualization

Business Intelligence tools transform analyzed data into meaningful visual information. **Power BI** is used in this project to create an interactive dashboard containing KPIs and graphical visualizations for customer and business analysis.

CHAPTER 3

DATASET AND PROJECT METHODOLOGY

3.1 Introduction

This chapter presents the dataset and methodology used for the **Customer Behavior Analysis** project. It explains the data source, tools and technologies, data preprocessing, exploratory data analysis, SQL-based analysis, churn analysis, business insight generation, and Power BI dashboard development.

3.2 Dataset Description

The project uses an **e-commerce customer dataset** containing information related to customers, transactions, products, payments, and purchasing activities.

The dataset is used to analyze:

- Customer characteristics
- Purchasing behavior
- Product categories
- Payment methods
- Orders and revenue
- Customer churn
- Business performance

Before performing the analysis, the dataset is inspected to understand its structure, columns, data types, missing values, duplicate records, and categorical values.

3.3 Tools and Technologies Used

The project uses multiple technologies, with each tool performing a specific role in the analytical workflow.

3.3.1 Python

Python is used for **data preprocessing, data cleaning, exploratory data analysis, and statistical analysis**.

3.3.2 Python Libraries

Various Python libraries are used according to the requirements of the analysis. These libraries support:

- Data manipulation
- Numerical calculations
- Data visualization
- Statistical analysis
- Data exploration

3.3.3 PostgreSQL and

PostgreSQL is used as the database environment for storing and querying the cleaned dataset. SQL is mainly used for:

- Customer analysis
- Order and revenue analysis
- **Churn analysis**
- **Business insight generation**

3.3.4 Power BI

Power BI is used to create the final **interactive customer behavior dashboard**. It presents KPIs, charts, and analytical results in an easy-to-understand format.

3.3.5 Jupyter Notebook

Jupyter Notebook is used as the working environment for Python-based data cleaning, analysis, and experimentation.

3.4 Project Methodology

The project follows a systematic analytical workflow:

Data Collection → Data Cleaning → EDA → SQL Analysis → Churn Analysis → Business Insights → Power BI Dashboard

Each stage contributes to converting raw customer data into useful business information.

3.4.1 Data Collection

The e-commerce customer dataset is collected and loaded into the Python environment for initial inspection and analysis.

3.4.2 Data Cleaning

The dataset is checked for missing values, duplicate records, incorrect data types, and inconsistent categorical values. Required preprocessing operations are performed to improve data quality.

3.4.3 Exploratory Data Analysis

EDA is performed using Python to understand the distribution of variables, customer characteristics, purchasing patterns, and relationships between different attributes.

3.4.4 SQL Analysis

The cleaned dataset is loaded into PostgreSQL. SQL queries are then used to perform structured analysis of customers, orders, revenue, and other business-related variables.

3.4.5 Churn Analysis

Customer churn is analyzed using SQL. The analysis helps identify customer loss patterns and provides information that can be used to understand customer retention.

3.4.6 Business Insights

The results obtained from SQL analysis are interpreted to identify important business patterns related to customers, orders, revenue, purchasing behavior, and churn.

3.4.7 Power BI Dashboard

The analytical results are presented through an interactive Power BI dashboard containing KPIs and different visualizations.

3.5 Data Preprocessing and Cleaning

Data preprocessing is performed before detailed analysis to ensure that the dataset is suitable for analytical operations.

The major preprocessing activities include:

3.5.1 Checking Missing Values

The dataset is examined for null or missing values. Missing values are identified and handled according to the requirements of the respective columns.

3.5.3 Checking Data Types

The data types of different columns are examined to ensure that numerical, categorical, and date-related fields are stored correctly.

3.5.4 Checking Categorical Values

Categorical variables such as **product category, payment method, and gender** are examined for inconsistent or unexpected values.

3.6 Exploratory Data Analysis

Exploratory Data Analysis is performed using Python to obtain an initial understanding of the dataset.

The analysis focuses on:

- Distribution of customer-related variables
- Product category patterns
- Payment method distribution
- Purchasing behavior
- Numerical variable distributions
- Relationships between important variables
- Identification of unusual observations

EDA helps in understanding the structure of the data and provides a foundation for further SQL-based analysis.

3.7 SQL-Based Analysis

After data preprocessing and EDA, the cleaned dataset is imported into PostgreSQL.

SQL queries are used to perform structured analysis using operations such as filtering, grouping, aggregation, sorting, and conditional analysis.

The SQL analysis focuses on:

1. **Customer Analysis**
2. **Order Analysis**
3. **Revenue Analysis**
4. **Churn Analysis**
5. **Business Insight Generation**

This stage converts the prepared data into meaningful analytical results.

3.8 Customer Churn Analysis

Customer churn analysis is one of the major components of the project.

SQL is used to analyze customer records and identify churn-related patterns. The analysis helps determine the proportion and characteristics of customers who are classified as churned.

The churn analysis is used to understand:

- Overall churn rate
- Customer retention patterns
- Characteristics of churned customers
- Relationship between customer behavior and churn

The results can help businesses develop appropriate customer-retention strategies.

3.9 Business Insight Generation

Business insights are derived primarily from the **SQL analysis** of customer and transaction data.

The analysis focuses on identifying meaningful patterns related to:

- Customer behavior
- Revenue generation
- Order activity
- Product categories
- Payment methods
- Customer churn.

3.10 Power BI Dashboard Development

The dashboard contains important Key Performance Indicators (KPIs), including:

- 👥 **Total Customers**
- 💰 **Total Revenue**
- 🛒 **Total Orders**
- 📉 **Churn Rate**

Different visualizations are used to represent customer and business information, including:

- **Bar Chart** – Category and comparative analysis
- **Area Chart** – Trend analysis
- **Scatter Plot** – Relationship analysis between numerical variables
- **KPI Cards** – High-level business metrics

CHAPTER 4

DATA PREPROCESSING AND EDA

4.1 Introduction

Data preprocessing is an important step in the Customer Behavior Analysis project. It ensures that the dataset is clean, consistent, and suitable for further analysis. After preprocessing, Exploratory Data Analysis (EDA) is performed to understand the major patterns and characteristics of the data.

4.2 Data Inspection

The dataset was initially inspected using Python to understand its structure, number of records, columns, data types, and overall data quality.

The initial inspection helped identify the variables required for further analysis and determined the preprocessing operations needed.

```
[12]: df.head()
```

[12]:

	Customer ID	Purchase Date	Product Category	Product Price	Quantity	Total Purchase Amount	Payment Method	Customer Age	Returns	Customer Name	Age	Gender	Churn
0	44605	2023-05-03 21:30:02	Home	177	1	2427	PayPal	31	1.0	John Rivera	31	Female	0
1	44605	2021-05-16 13:57:44	Electronics	174	3	2448	PayPal	31	1.0	John Rivera	31	Female	0
2	44605	2020-07-13 06:16:57	Books	413	1	2345	Credit Card	31	1.0	John Rivera	31	Female	0
3	44605	2023-01-17 13:14:36	Electronics	396	3	937	Cash	31	0.0	John Rivera	31	Female	0
4	44605	2021-05-01 11:29:27	Books	259	4	2598	PayPal	31	1.0	John Rivera	31	Female	0

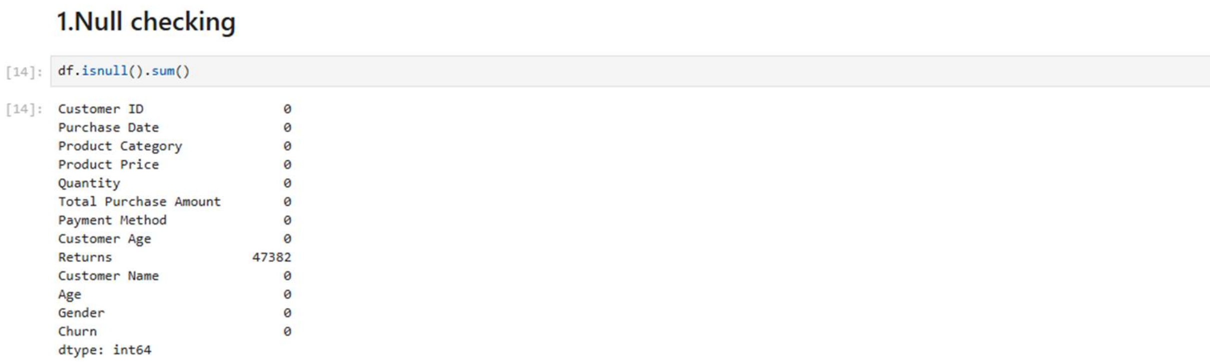
```
[13]: df.info()

<class 'pandas.DataFrame'>
RangeIndex: 250000 entries, 0 to 249999
Data columns (total 13 columns):
 #   Column              Non-Null Count   Dtype
---  ------              --------------   -----
 0   Customer ID         250000 non-null  int64
 1   Purchase Date       250000 non-null  str
 2   Product Category    250000 non-null  str
 3   Product Price       250000 non-null  int64
```

Figure 4.1: Initial Dataset Inspection

4.3 Missing Value Analysis

The dataset was checked for missing or null values using Python. This step was performed to ensure that missing information did not negatively affect the analysis.

The missing values were identified and handled appropriately during the data-cleaning process.



1.Null checking

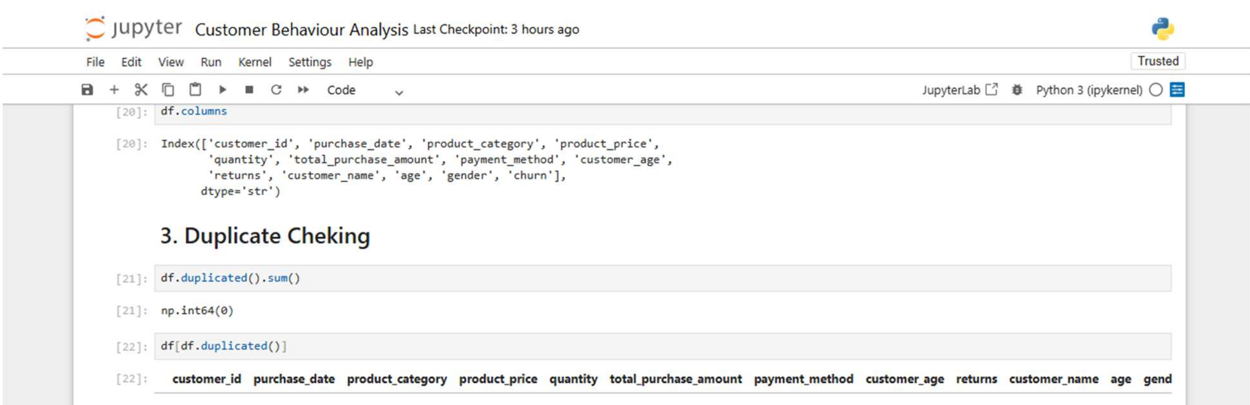
```
[14]: df.isnull().sum()

[14]: Customer ID                 0
      Purchase Date               0
      Product Category            0
      Product Price               0
      Quantity                    0
      Total Purchase Amount       0
      Payment Method              0
      Customer Age                0
      Returns                 47382
      Customer Name               0
      Age                         0
      Gender                      0
      Churn                       0
      dtype: int64
```

Figure 4.2: Missing Value Analysis

4.4 Duplicate Record Analysis

Duplicate records were checked to ensure that the same customer or transaction was not unnecessarily repeated in the dataset.

Removing duplicate records helps maintain the accuracy of calculations and prevents repeated observations from affecting the analysis.




```
[20]: df.columns

[20]: Index(['customer_id', 'purchase_date', 'product_category', 'product_price',
             'quantity', 'total_purchase_amount', 'payment_method', 'customer_age',
             'returns', 'customer_name', 'age', 'gender', 'churn'],
            dtype='str')
```

3. Duplicate Cheking

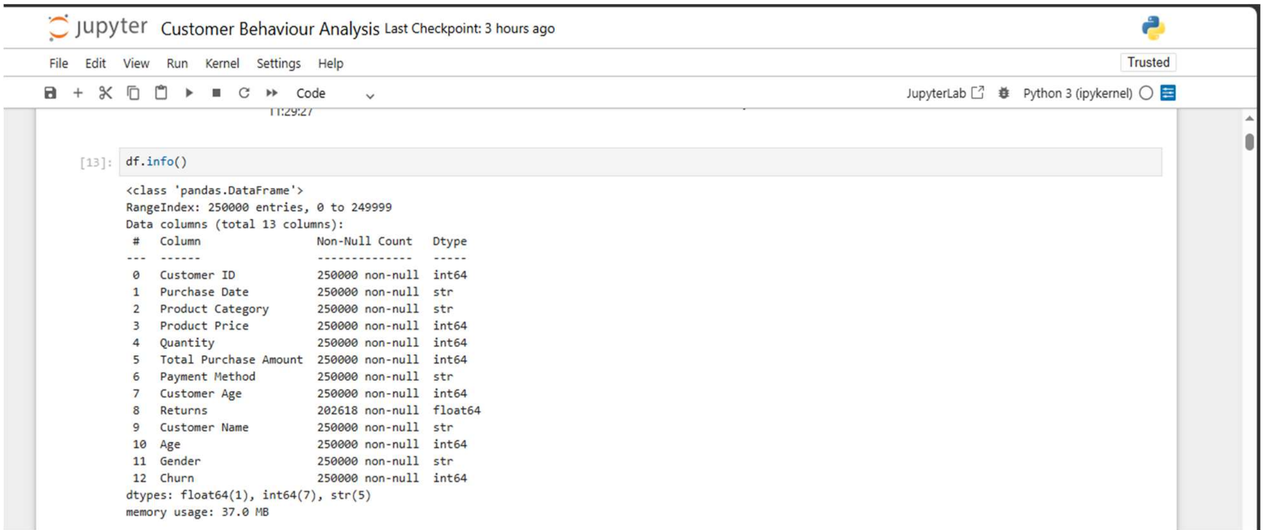
```
[21]: df.duplicated().sum()

[21]: np.int64(0)

[22]: df[df.duplicated()]

[22]: customer_id purchase_date product_category product_price quantity total_purchase_amount payment_method customer_age returns customer_name age gend
```

Figure 4.3: Duplicate Record Analysis

4.5 Data Type Validation

The data types of the dataset columns were examined to ensure that numerical, categorical, and other variables were stored correctly.

Correct data types are necessary for performing mathematical calculations, filtering, grouping, and further analysis.




```
<class 'pandas.DataFrame'>
RangeIndex: 250000 entries, 0 to 249999
Data columns (total 13 columns):
 #   Column                 Non-Null Count   Dtype
---  ------                 --------------   -----
 0   Customer ID            250000 non-null  int64
 1   Purchase Date          250000 non-null  str
 2   Product Category       250000 non-null  str
 3   Product Price          250000 non-null  int64
 4   Quantity               250000 non-null  int64
 5   Total Purchase Amount  250000 non-null  int64
 6   Payment Method         250000 non-null  str
 7   Customer Age           250000 non-null  int64
 8   Returns                202618 non-null  float64
 9   Customer Name          250000 non-null  str
 10  Age                    250000 non-null  int64
 11  Gender                 250000 non-null  str
 12  Churn                  250000 non-null  int64
dtypes: float64(1), int64(7), str(5)
memory usage: 37.0 MB
```

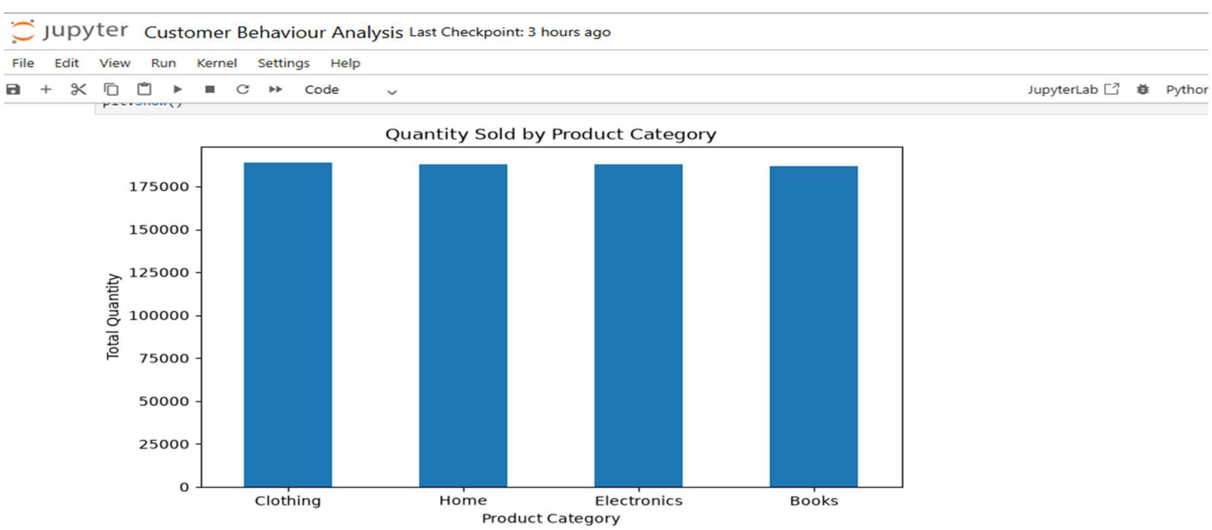
Figure 4.4: Data Type Validation

4.7 Exploratory Data Analysis

After data cleaning, EDA was performed using Python to understand the dataset and identify important patterns.

The analysis included:

- Customer characteristics
- Product category distribution
- Payment method distribution
- Numerical variable distributions
- Customer purchasing patterns
- Relationships between important variables



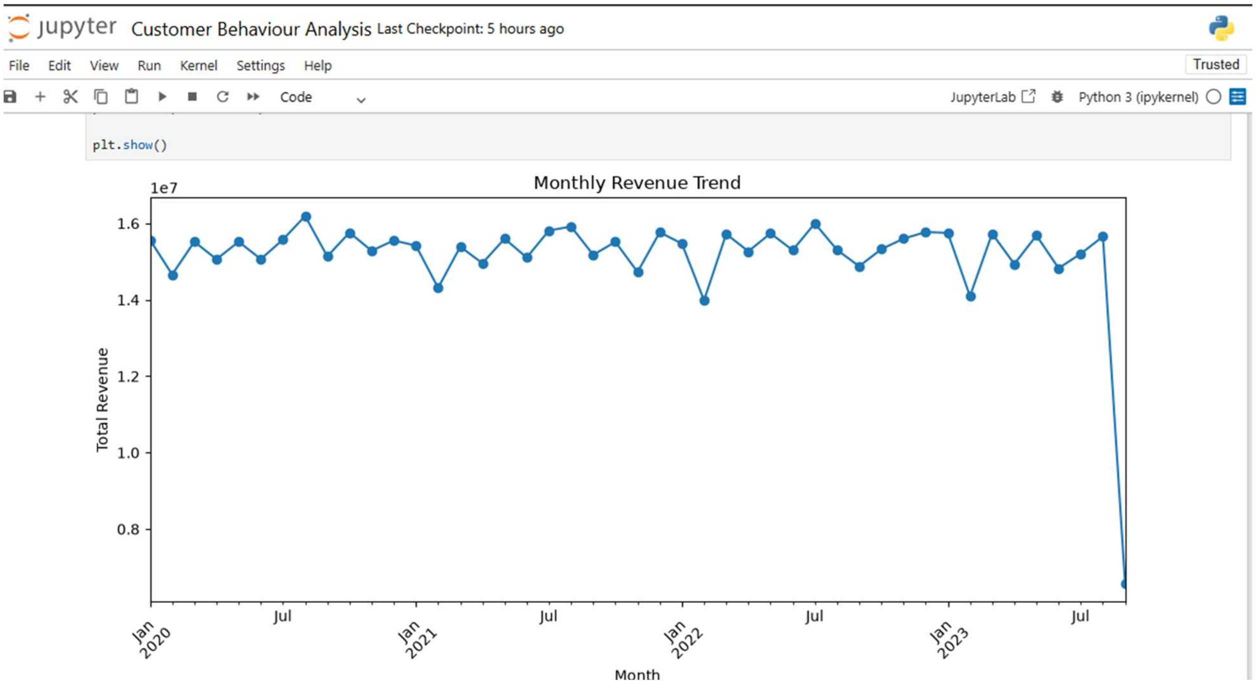
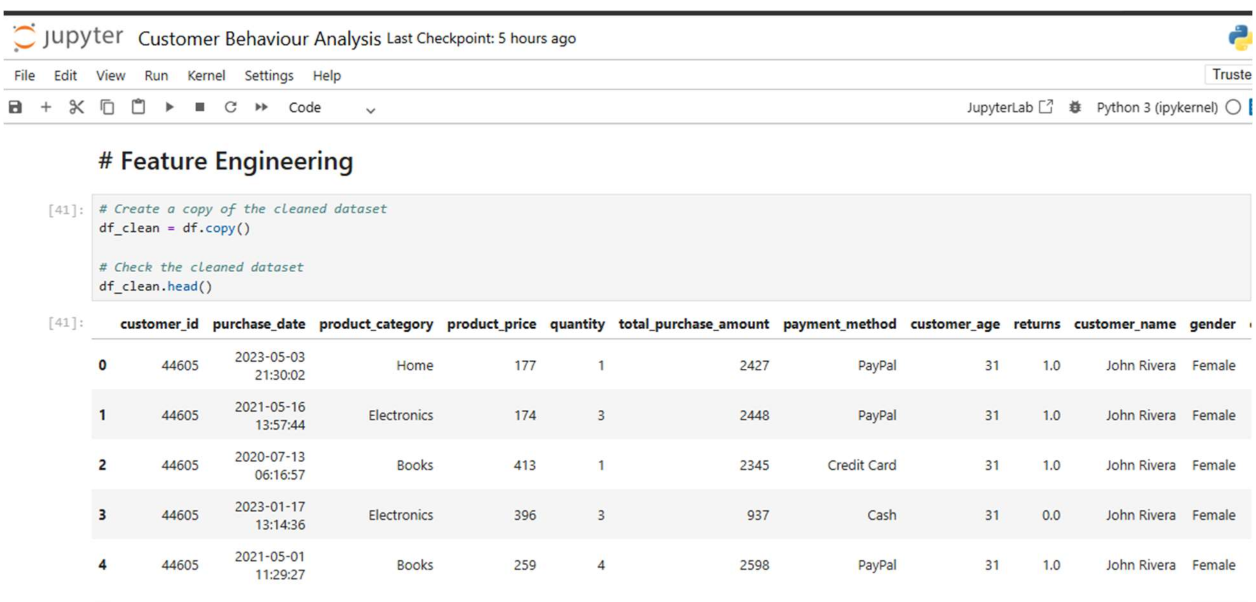


Figure 4.6: Exploratory Data Analysis

4.8 Feature Engineering

Feature engineering is the process of creating or transforming variables to make the dataset more useful for analysis. In this project, relevant customer and transaction attributes were prepared to support customer behavior, revenue, order, and churn analysis.



Feature Engineering

[41]:
```
# Create a copy of the cleaned dataset
df_clean = df.copy()

# Check the cleaned dataset
df_clean.head()
```

[41]:

	customer_id	purchase_date	product_category	product_price	quantity	total_purchase_amount	payment_method	customer_age	returns	customer_name	gender
0	44605	2023-05-03 21:30:02	Home	177	1	2427	PayPal	31	1.0	John Rivera	Female
1	44605	2021-05-16 13:57:44	Electronics	174	3	2448	PayPal	31	1.0	John Rivera	Female
2	44605	2020-07-13 06:16:57	Books	413	1	2345	Credit Card	31	1.0	John Rivera	Female
3	44605	2023-01-17 13:14:36	Electronics	396	3	937	Cash	31	0.0	John Rivera	Female
4	44605	2021-05-01 11:29:27	Books	259	4	2598	PayPal	31	1.0	John Rivera	Female

CHAPTER 5

CHURN ANALYSIS AND BUSINESS INSIGHTS

5.1 Introduction

After data cleaning and exploratory analysis, the cleaned dataset was loaded into **PostgreSQL** for further analysis. SQL queries were used to analyze customer behavior, orders, revenue, churn, and business performance.

5.2 Database Setup

The cleaned dataset was imported into PostgreSQL and stored in a structured table. The database was then used to perform analytical queries on customer and transaction data.

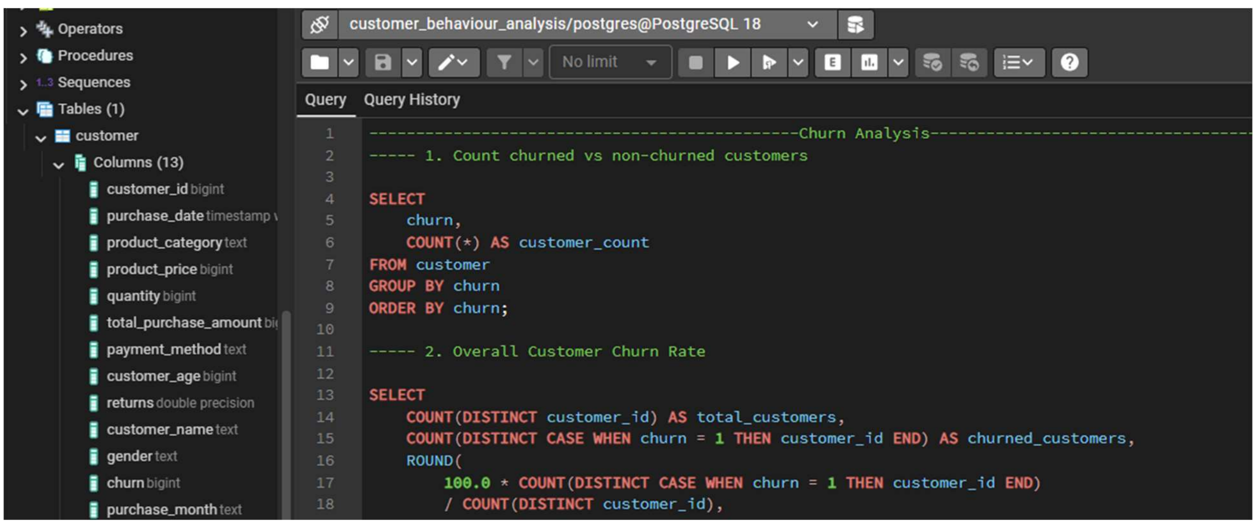


Figure 5.1: Customer Dataset in PostgreSQL

5.3 Overall Customer Churn Rate

The overall customer churn rate was calculated using SQL by comparing the number of churned customers with the total number of customers.

The churn rate provides an overall measure of customer loss and helps evaluate customer retention.

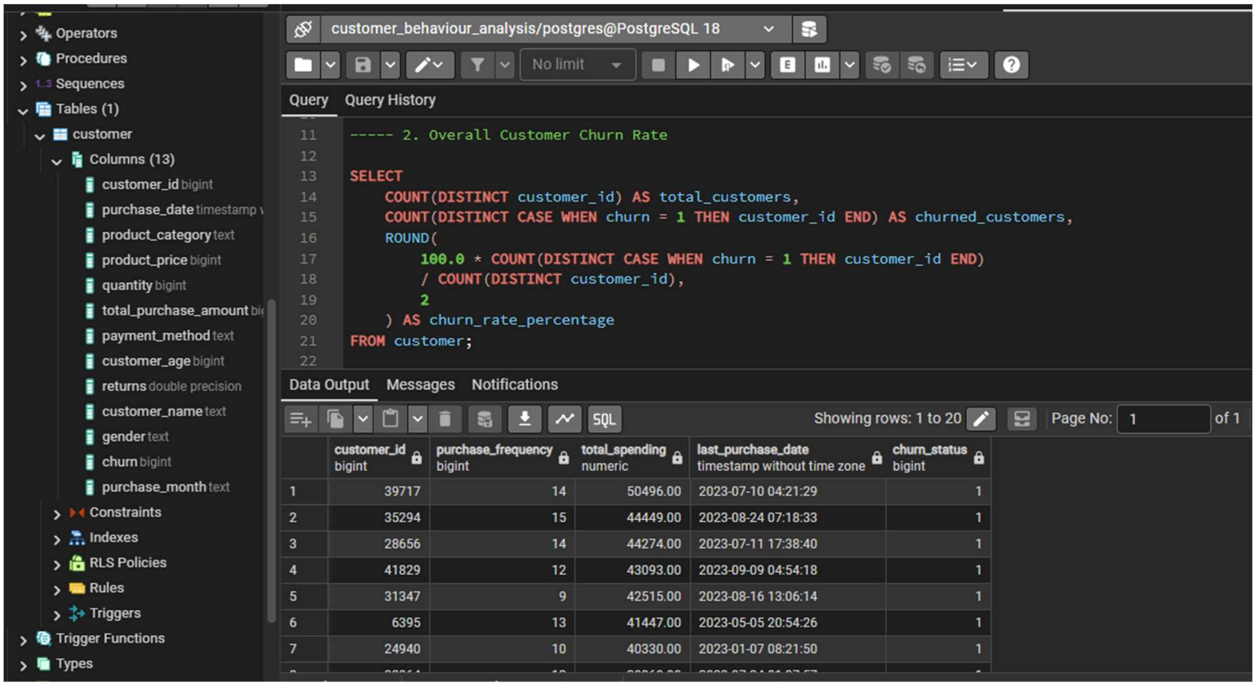


Figure 5.2 : Overall Customer Churn Rate

5.4 Churn Rate by Gender

Churn rate was further analyzed by gender to compare customer churn behavior across different gender groups.

This analysis helps determine whether there is a noticeable difference in churn rates between the groups.

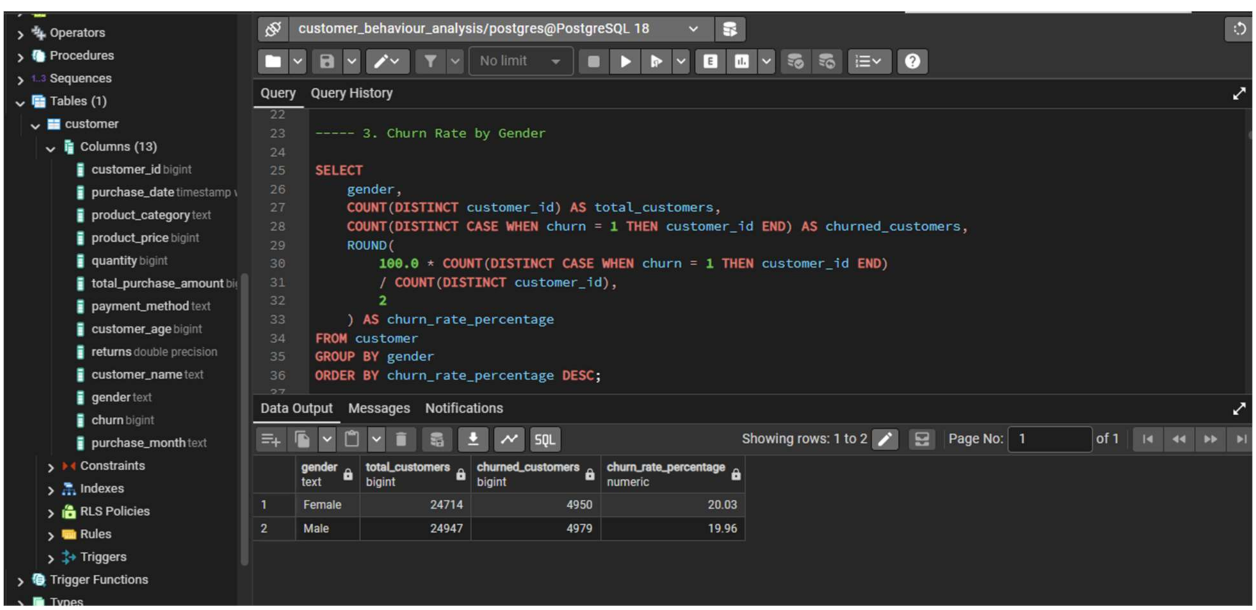


Figure 5.3 : Churn Rate by Gender

5.5 Monthly Revenue Trend

Monthly revenue was analyzed using SQL to understand how revenue changes over time.

The monthly trend helps identify periods of higher or lower revenue and provides an overview of the business's revenue performance over the selected period.

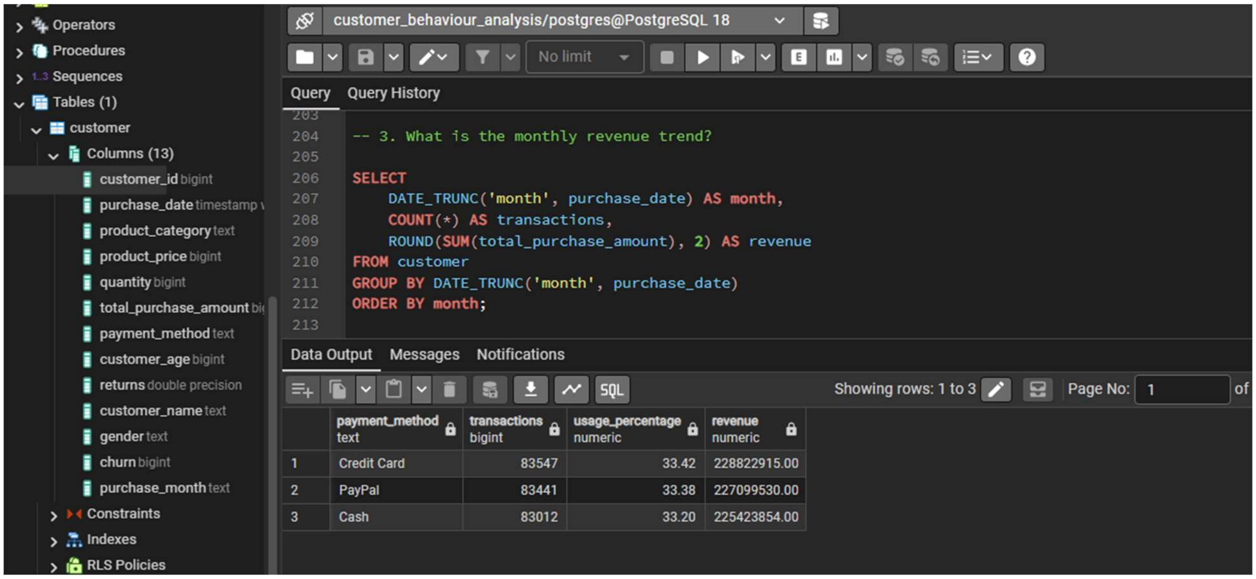


Figure 5.4 : Monthly Revenue Trend

5.4 Business Insights Using SQL

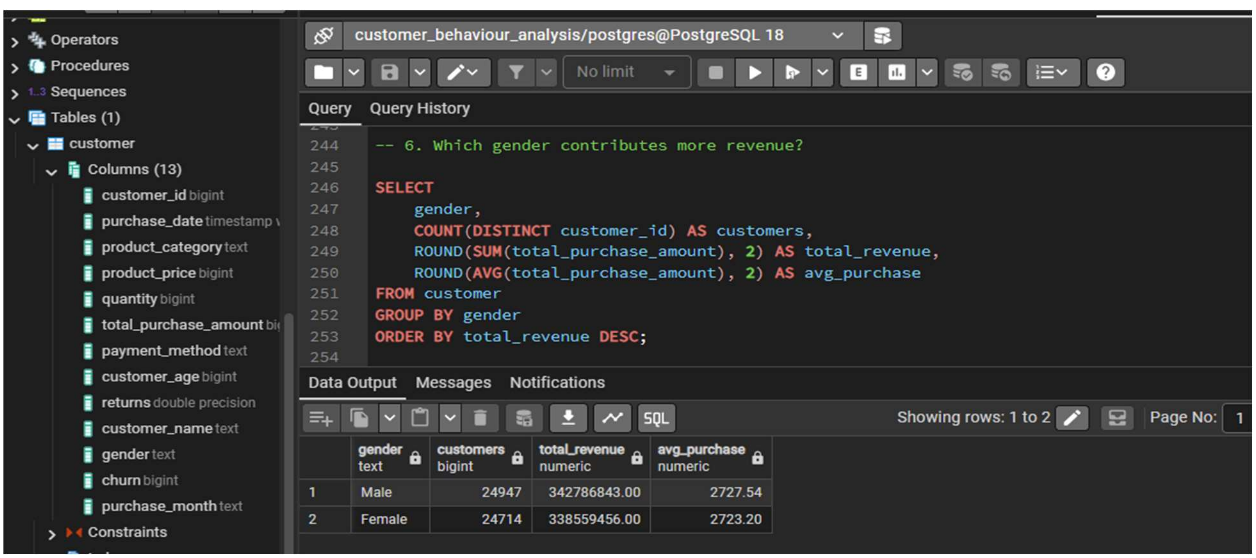


Figure 5.7: Gender Wise Contribution

CHAPTER 6

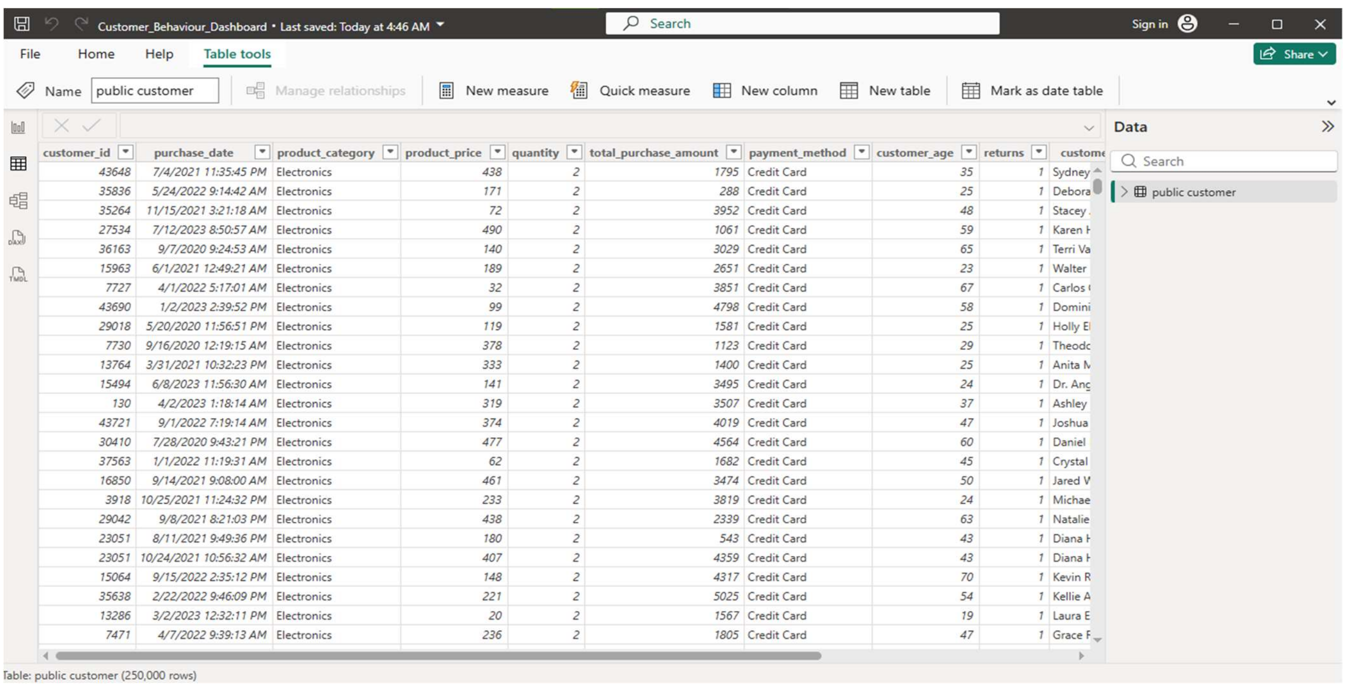
POWER BI DASHBOARD AND DATA VISUALIZATION

6.1 Introduction

Power BI is used in this project to present the results of the customer behavior analysis in an interactive and easy-to-understand format. The dashboard combines important KPIs and visualizations to provide an overall view of customer and business performance.

6.2 Data Integration

The analyzed data is connected to Power BI for visualization. The data is organized and prepared so that the required customer, order, revenue, and churn information can be represented through different visuals.



customer_id	purchase_date	product_category	product_price	quantity	total_purchase_amount	payment_method	customer_age	returns	custome
43648	7/4/2021 11:35:45 PM	Electronics	438	2	1795	Credit Card	35	1	Sydney
35836	5/24/2022 9:14:42 AM	Electronics	171	2	288	Credit Card	25	1	Debora
35264	11/15/2021 3:21:18 AM	Electronics	72	2	3952	Credit Card	48	1	Stacey
27534	7/12/2023 8:50:57 AM	Electronics	490	2	1061	Credit Card	59	1	Karen H
36163	9/7/2020 9:24:53 AM	Electronics	140	2	3029	Credit Card	65	1	Terri Va
15963	6/1/2021 12:49:21 AM	Electronics	189	2	2651	Credit Card	23	1	Walter
7727	4/1/2022 5:17:01 AM	Electronics	32	2	3851	Credit Card	67	1	Carlos
43690	1/2/2023 2:39:52 PM	Electronics	99	2	4798	Credit Card	58	1	Domini
29018	5/20/2020 11:56:51 PM	Electronics	119	2	1581	Credit Card	25	1	Holly E
7730	9/16/2020 12:19:15 AM	Electronics	378	2	1123	Credit Card	29	1	Theodc
13764	3/31/2021 10:32:23 PM	Electronics	333	2	1400	Credit Card	25	1	Anita M
15494	6/8/2023 11:56:30 AM	Electronics	141	2	3495	Credit Card	24	1	Dr. Ang
130	4/2/2023 1:18:14 AM	Electronics	319	2	3507	Credit Card	37	1	Ashley
43721	9/1/2022 7:19:14 AM	Electronics	374	2	4019	Credit Card	47	1	Joshua
30410	7/28/2020 9:43:21 PM	Electronics	477	2	4564	Credit Card	60	1	Daniel
37563	1/1/2022 11:19:31 AM	Electronics	62	2	1682	Credit Card	45	1	Crystal
16850	9/14/2021 9:08:00 AM	Electronics	461	2	3474	Credit Card	50	1	Jared W
3918	10/25/2021 11:24:32 PM	Electronics	233	2	3819	Credit Card	24	1	Michae
29042	9/8/2021 8:21:03 PM	Electronics	438	2	2339	Credit Card	63	1	Natalie
23051	8/11/2021 9:49:36 PM	Electronics	180	2	543	Credit Card	43	1	Diana H
23051	10/24/2021 10:56:32 AM	Electronics	407	2	4359	Credit Card	43	1	Diana H
15064	9/15/2022 2:35:12 PM	Electronics	148	2	4317	Credit Card	70	1	Kevin R
35638	2/22/2022 9:46:09 PM	Electronics	221	2	5025	Credit Card	54	1	Kellie A
13286	3/2/2023 12:32:11 PM	Electronics	20	2	1567	Credit Card	19	1	Laura E
7471	4/7/2022 9:39:13 AM	Electronics	236	2	1805	Credit Card	47	1	Grace F

Table: public customer (250,000 rows)

Figure 6.1: Data Integration in Power BI

6.3 Dashboard Design

The dashboard is designed to provide a quick overview of the major business metrics. KPI cards are placed prominently at the top, followed by graphical visualizations for detailed analysis.

The dashboard focuses on:

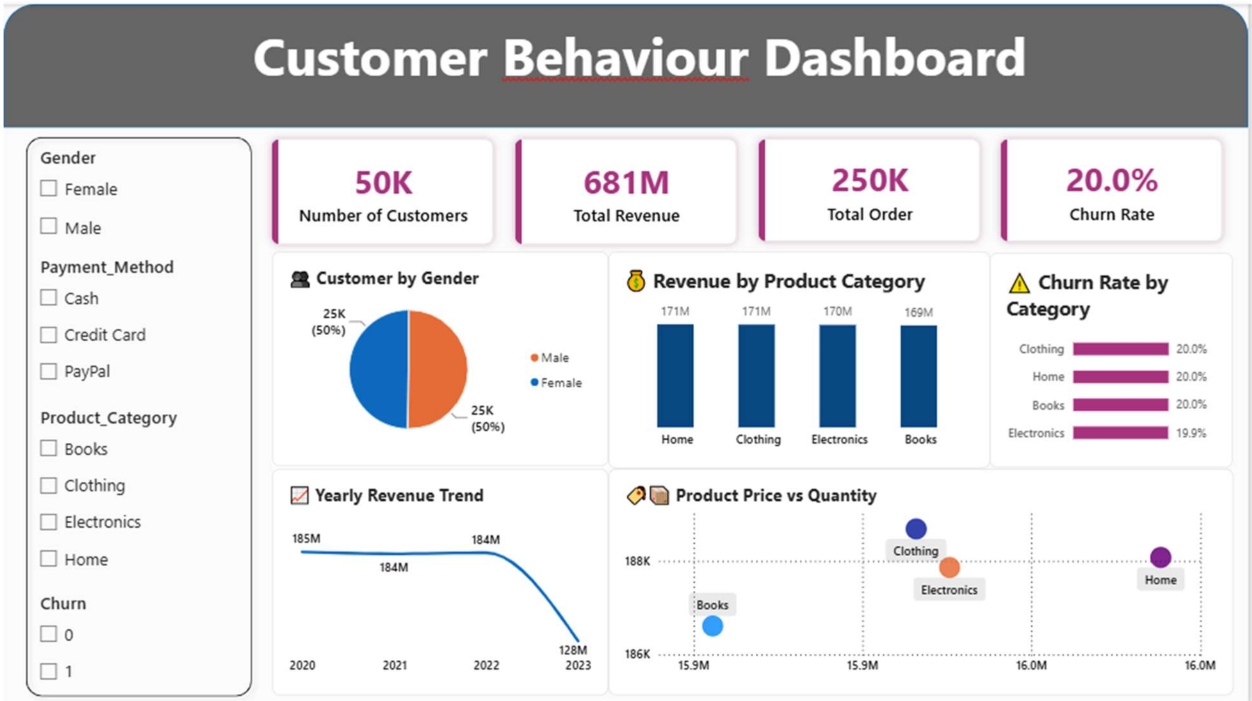


Figure 6.2: Customer Behavior Analysis Dashboard

6.4 Key Performance Indicators

The dashboard contains four major KPIs:

KPI	Purpose
👥 **Total Customers**	Shows the total customer base
💰 **Total Revenue**	Shows the total revenue generated
🛒 **Total Orders**	Shows total order activity
📉 **Churn Rate**	Shows the percentage of churned customers

6.11 Dashboard Summary

The Power BI dashboard combines the analytical results into a single interactive interface. It provides a clear view of customers, revenue, orders, and churn while allowing users to explore the underlying patterns through different visualizations.

CHAPTER 7

RESULTS

7.1 Introduction

This chapter presents the major results obtained from the Customer Behavior Analysis project. The results are based on the analysis performed using Python, SQL, and Power BI.

7.2 Key Findings

The analysis provides an overall understanding of customer activity, orders, revenue, purchasing behavior, and churn.

The major findings include:

- The total customer base and order activity were identified.
- Revenue performance was analyzed across relevant categories.
- Customer purchasing patterns were examined.
- Customer churn was analyzed using SQL.
- Important business insights were derived from SQL analysis.
- The results were presented through an interactive Power BI dashboard.

7.3 Customer Insights

Customer analysis helps identify differences in customer behavior and purchasing activity. The results can be used to understand customer groups and their contribution to overall business performance.

7.4 Revenue Insights

Revenue analysis provides information about the financial performance of the business. Comparing revenue across relevant categories helps identify areas that contribute significantly to total revenue.

7.5 Order Insights

Order analysis provides an overview of purchasing activity. The total number of orders and their distribution help in understanding customer transaction behavior.

7.6 Churn Insights

Churn analysis performed using SQL provides information about customer loss and retention. The churn rate helps measure the proportion of customers who are classified as churned.

The results can help identify areas where customer-retention strategies may be required.

7.7 Business Insights

The SQL analysis provides several business-oriented insights from customer and transaction data. These insights can help businesses understand customer activity, revenue generation, order patterns, and churn.

The findings can support decisions related to:

- Customer retention
- Customer targeting
- Revenue improvement
- Product performance
- Customer engagement

7.8 Business Recommendations

Based on the analysis, businesses can consider the following recommendations:

1. **Improve customer retention** by identifying and engaging churn-prone customers.
2. **Focus on high-value customers** who contribute significantly to revenue.
3. **Analyze purchasing patterns** to develop targeted marketing strategies.
4. **Monitor revenue and order trends** regularly to identify changes in business performance.
5. **Use customer insights** to improve personalized offers and customer engagement.

7.10 Chapter Summary

This chapter presented the key results and business insights obtained from the project. SQL-based analysis was used for churn and business insight generation, while Power BI was used to present the results visually. The findings provide a clear understanding of customer behavior and overall business performance.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The **Customer Behavior Analysis** project successfully demonstrates the use of data analytics techniques to understand customer and business performance.

Python was used for data preprocessing and exploratory data analysis, while **SQL was used for customer churn analysis and business insight generation**. Power BI was then used to present the results through an interactive dashboard containing important KPIs and visualizations.

The project provides a structured approach for converting raw customer data into meaningful information that can support data-driven business decisions.

8.2 Project Outcomes

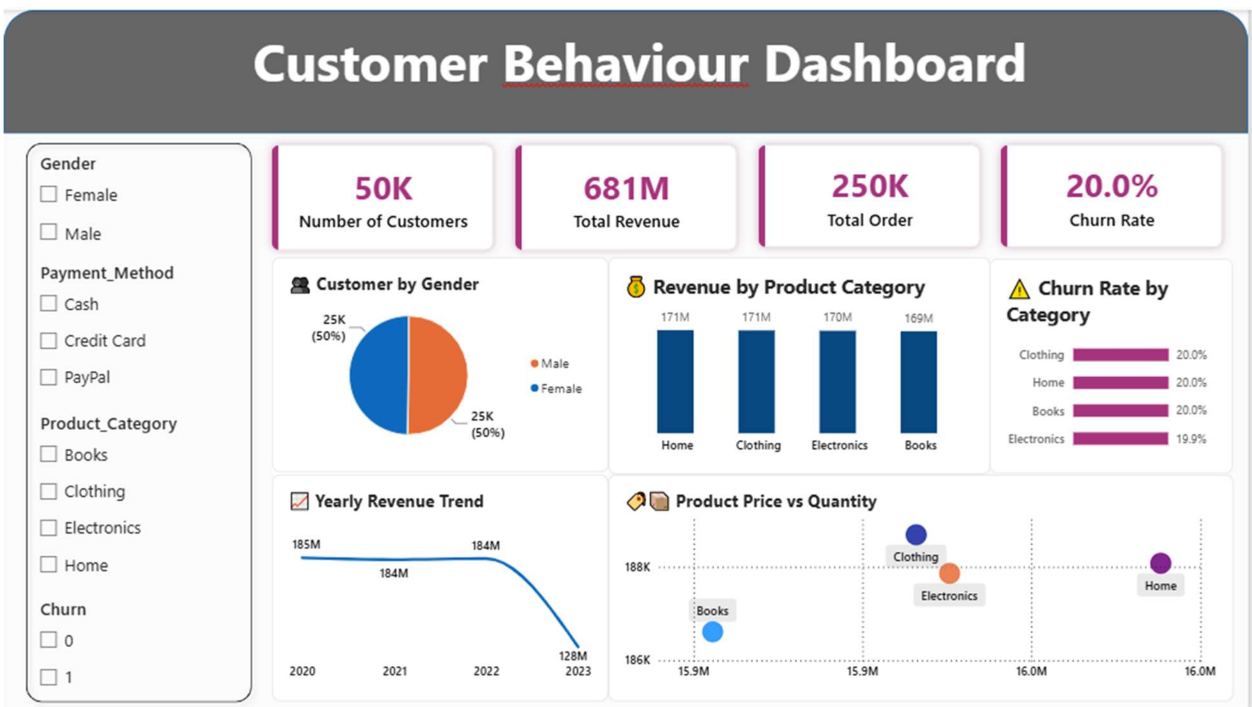


Figure 8.1: Final Project Outcome

The major outcomes of the project are:

- Cleaned and prepared customer data.
- Performed exploratory data analysis using Python.

- Conducted customer, order, and revenue analysis using SQL.
- Performed churn analysis using SQL.
- Generated business insights from SQL analysis.
- Developed an interactive Power BI dashboard.
- Visualized important KPIs such as total customers, revenue, orders, and churn rate.

8.3 Limitations

The project has some limitations:

- The analysis depends on the quality and availability of the dataset.
- The results are based on historical customer and transaction data.
- Churn analysis is based on the available customer attributes and defined churn criteria.
- The project focuses on descriptive and analytical insights rather than predictive churn modeling.

8.4 Future Scope

The project can be further enhanced by incorporating advanced analytics and machine learning techniques.

Future improvements may include:

1. **Churn Prediction** – Develop a machine learning model to predict customers who are likely to churn.
2. **Advanced Customer Segmentation** – Apply clustering algorithms such as K-Means.
3. **Real-Time Dashboard** – Connect Power BI with continuously updated data.
4. **Predictive Revenue Analysis** – Forecast future revenue and customer purchasing activity.
5. **Recommendation System** – Recommend products based on individual customer behavior.
6. **Automated Reporting** – Generate periodic customer and business performance reports automatically.

8.5 Final Summary

The project demonstrates how **Python, SQL, and Power BI** can be integrated into a complete customer analytics workflow. The analysis provides useful information about customer behavior, business performance, and churn, while the interactive dashboard makes the results easier to interpret and communicate.

Overall, the project provides a practical foundation for applying data analytics to customer behavior and supporting data-driven business decision-making.

CHAPTER 9

REFERENCES AND APPENDIX

1. Python Software Foundation. **Python Documentation.** https://docs.python.org/
2. Pandas Development Team. **Pandas Documentation.** https://pandas.pydata.org/docs/
3. NumPy Developers. **NumPy Documentation.** https://numpy.org/doc/
4. Matplotlib Development Team. **Matplotlib Documentation.** https://matplotlib.org/stable/
5. Seaborn Development Team. **Seaborn Documentation.** https://seaborn.pydata.org/
6. SciPy Developers. **SciPy Documentation.** https://docs.scipy.org/doc/scipy/
7. PostgreSQL Global Development Group. **PostgreSQL Documentation.** https://www.postgresql.org/docs/
8. Microsoft. **Microsoft Power BI Documentation.** https://learn.microsoft.com/power-bi/
9. SQLAlchemy Documentation. **SQLAlchemy Documentation.** https://docs.sqlalchemy.org/
10. Project Dataset. **E-Commerce Customer Dataset.** https://www.kaggle.com/datasets/bhanupratapbiswas/customer-behavior-analysis

9.2 Appendix

The appendix contains supporting materials related to the implementation of the project that are not included in the main chapters.

The appendix may include:

- Python code used for data preprocessing and EDA.
- Important SQL queries used for customer and transaction analysis.

9.4 Final Project Structure

The complete project follows the sequence:

Dataset → Python Data Cleaning → Exploratory Data Analysis → PostgreSQL → SQL Analysis → Churn Analysis → Business Insights → Power BI Dashboard → Results and Recommendations